

Evidence-Gated Residual Doppler Correction for LR-FHSS Direct-to-Satellite IoT

Anonymous Author 1
Affiliation omitted for
double-blind review
Email omitted

Anonymous Author 2
Affiliation omitted for
double-blind review
Email omitted

Anonymous Author 3
Affiliation omitted for
double-blind review
Email omitted

Abstract—Direct-to-Satellite (D2S) IoT terminals pre-compensate LEO Doppler for every LR-FHSS uplink using stale orbital elements propagated by SGP4. Learning a residual correction on top of that baseline raises a deployment question rather than a modelling one: a measurable prediction gain need not be reliable, nor operationally useful. We present an evidence-gated residual correction framework: inter-TLE residuals are turned into a supervised target, a low-capacity candidate family is fitted, and the learned branch is admitted only when a strictly later validation window improves on the physics baseline by a predefined margin. Using model-derived Space-Track element histories for nine LEO objects spanning five orbital regimes and six staleness bands from 8 to 168 h, the gate opens in none of 54 target-specific configurations, and limited BLACK KITE cross-direction checks likewise remain closed. A five-threshold residual-screening study yields exactly one opening among 270 configurations, and that opening is not robust to adjacent thresholds. Residual structure is nevertheless detectable: an IRIDIUM case shows a reproducible $\approx 1.9\%$ held-out improvement that survives multiplicity and temporal diagnostics. Its absolute size is 3.3 mHz, against a 500 Hz control tolerance that lies 917 residual standard deviations away, and it yields no resolvable endpoint-budget benefit. These results separate statistical detectability, deployment margin and endpoint value. All references are later-TLE SGP4 propagations rather than measured Doppler; no packet-level or over-the-air performance is claimed.

Index Terms—Direct-to-Satellite IoT, LR-FHSS, endpoint control, stale TLE, SGP4, inter-TLE residual, validation-gated deployment, LEO satellite

I. Introduction

Direct-to-Satellite (D2S) IoT over Low-Earth Orbit (LEO) connects power-constrained end devices without terrestrial infrastructure, but the high relative velocity induces Doppler shifts of order 20 kHz at 868 MHz and rapid Doppler-rate changes that stress the physical layer [1]. Long-Range Frequency-Hopping Spread Spectrum (LR-FHSS) [2] spreads transmit energy across narrow frequency bins, so residual frequency error after pre-compensation widens the guard the terminal must reserve, raises the chance of missing a hop bin, and spends airtime a battery-powered node cannot recover.

The terminal must size these margins before transmitting, from a stale TLE propagated open-loop by SGP4/SDP4 [3]. That baseline is already strong: the inter-TLE frequency residual it leaves is well under one Hz at short staleness. Whether a learned corrector improves on it is usually settled by measuring a prediction gain. We

argue that this is the wrong test, and that three criteria must be kept apart:

$$\underbrace{\text{detectability}}_{\text{is the gain real?}} \neq \underbrace{\text{margin}}_{\text{large enough?}} \neq \underbrace{\text{endpoint value}}_{\text{changes the budget?}} \quad (1)$$

A corrector can satisfy the first and fail the other two, and deploying on the first alone is what an always-on learned branch does. We therefore study the pre-transmission decision directly, formalised as an Evidence Gate: a learned residual is admitted only when a chronological, leakage-free validation window beats the physics baseline by a predefined margin γ , and SGP4 is retained otherwise, on real Space-Track element histories for nine LEO objects across five orbital regimes.

Related work. Prior art spans link, capacity and energy analyses of aggregate LR-FHSS behaviour under LEO Doppler and load [1], [2], [4]; synchronisation and pre-compensation, from NTN timing advance and ToA/CFO estimation [5] to endpoint Doppler pre-compensation driven by a satellite-broadcast beacon [6]; and orbit learning, which corrects SGP4/SDP4 [3] trajectories for trajectory accuracy [7]–[9], while D2S uplink policies [10] choose when to transmit given a prediction assumed trustworthy. The closest, [6], also pre-compensates at the end device but estimates the shift from a downlink beacon; we assume no downlink and correct open-loop from a stale element set, so our question is not how to estimate the shift but whether a learned correction to it may be trusted. None decides, before transmission, whether a learned residual should be trusted. Contributions:

- Evidence-gated residual correction framework. A software method that constructs inter-TLE residual targets, fits a nested candidate family, and admits the learned branch only when a chronological validation margin is met, retaining SGP4 otherwise (Sec. II).
- Multi-regime real-data evaluation. Across nine LEO objects, five regimes and six staleness bands, no branch earns deployment at the pre-specified operating point; a five-threshold screening analysis yields one isolated, non-robust opening (Secs. IV-A–IV-C).
- Learnability-to-value diagnosis. A reproducible $\approx 1.9\%$ IRIDIUM gain is 3.3 mHz of absolute correction with no resolvable endpoint benefit, separating the three

criteria of Eq. (1) (Sec. IV-D).

II. System Model and Evidence-Gated Correction

A. Doppler, Stale TLE and the Inter-TLE Residual

In a LEO pass the one-way Doppler is $f_D(t) = -v_r(t) f_c/c$, with v_r the radial velocity, and the residual carrier-frequency error after pre-compensation is

$$f_{\text{res}}(t) = f_D(t) - \hat{f}_D(t) + f_{\text{osc}}, \quad (2)$$

where $\hat{f}_D$ is the prediction and f_{osc} the oscillator offset. Equation (2) separates the orbital-prediction term from endpoint RF errors, which can dominate the absolute residual; only the TLE-driven component is isolated here. Reliable hop reception is modelled as $|f_{\text{res}}|$ staying small relative to a tolerance F_{tol} —a control proxy, not a standard-compliant receiver model.

The terminal holds a TLE older than the present epoch; propagating it with SGP4 gives the deployable prediction $f_{\text{phys}}(t)$. We define a model-derived reference $D_{\text{ref}}(t)$ as the SGP4 propagation of a later element set for the same object at the same UTC—a newer orbit solution, not an RF measurement. The residual

$$r(t) = D_{\text{ref}}(t) - f_{\text{phys}}(t) \quad (3)$$

is what a TLE-only terminal can observe by comparing element sets; it encodes mean-element updates, drag-term changes and fit differences. The reference is not available while the stale element is in use, so $r(t)$ is retrospective supervision: computable only once the later element is published, and used to train correctors applied to subsequent transmissions. A learned corrector forms $f_{\text{ml}}(t) = f_{\text{phys}}(t) + \hat{r}(x_t)$ from a feature vector x_t described in Sec. III-D.

B. The Evidence Gate

Let V be a chronological validation window, strictly later than the training span and strictly earlier than the held-out test span, so no future element set leaks into model selection. On V we measure the mean absolute frequency error of both branches, $\text{MAE}_{\text{phys}}(V)$ and $\text{MAE}_{\text{ml}}(V)$, and admit learning only on a proven margin,

$$G = \mathbf{1}[\text{MAE}_{\text{ml}}(V) < \gamma \text{MAE}_{\text{phys}}(V)], \quad \gamma \in (0, 1], \quad (4)$$

deploying $\hat{f} = Gf_{\text{ml}} + (1 - G)f_{\text{phys}}$. Figure 1 places this decision inside the surrounding software method; Algorithm 1 states the full procedure. Training fits the candidates; validation both selects the candidate and decides G ; the held-out segment reports the consequence of that already-frozen policy and never selects a model or decides the gate. By construction the gate gives a validation-window property, not a distribution-shift bound. We use $\gamma = 0.95$, a conservatism setting fixed in a version-controlled protocol before any result was inspected, and report sensitivity in Sec. IV-D.

The held-out set E is not queried at any point before line 9; no quantity derived from E enters $\arg\min$, the margin test, or hyper-parameter choice. This is checked mechanically (Sec. III-E).

Algorithm 1 Evidence-Gated Residual Deployment

Require: canonical history H ; staleness target τ ; learned candidate set $\mathcal{M}_L$; margin γ ; screen τ_s

- 1: $P \leftarrow \text{BuildPairs}(H, \tau)$ $\triangleright$ stale/reference pairs, K samples each
- 2: $P \leftarrow \text{Screen}(P, \tau_s)$ $\triangleright$ drop pairs with $\max |r| > \tau_s$
- 3: $T, V, E \leftarrow \text{ChronologicalSplit}(P)$ $\triangleright$ by reference epoch, 60/20/20
- 4: for all $m \in \mathcal{M}_L$ do
- 5: $\text{Fit}(m, T)$; $J_m \leftarrow \text{MAE}(m, V)$
- 6: end for
- 7: $m^* \leftarrow \arg\min_{m \in \mathcal{M}_L} J_m$ $\triangleright$ references never compete
- 8: $G \leftarrow \mathbf{1}[J_{m^*} < \gamma \text{MAE}_{\text{phys}}(V)]$
- 9: $\text{Freeze}(m^*, G)$ $\triangleright$ policy is now immutable
- 10: report $\text{Evaluate}(m^*, G, E)$ $\triangleright$ E is queried only here

C. When the Gate Is Re-Decided

Fitting and gating are refresh-time operations, not per-transmission ones (Fig. 1).

On a new element set (median inter-element gap 4.5–10.0 h here) the residual history is extended, $\mathcal{M}_L$ is refitted on training, and m^* and G are re-decided on a fresh validation window; only the corrector, the gate bit and its metadata are persisted.

Between refreshes the state is immutable: a transmission applies $\hat{f} = f_{\text{phys}} + G\hat{r}_{m^*}(x)$ with no refitting, selection or gate evaluation. Continuously adapting the corrector would reintroduce the unaudited, always-on behaviour the gate exists to prevent and would make the acting policy impossible to attribute afterwards. When evidence lapses the gate reverts to $G = 0$.

III. Data, Software Pipeline and Evaluation Protocol

A. Dataset Qualification

The acquisition and qualification contract was fixed in a version-controlled protocol before evaluation and executed before any learner was run. A satellite is retained only if at least $B_{\text{min}} = 2$ staleness bands are supported, where a band counts as supported only when the chronological split yields at least $n_{\text{min}} = 3$ pairs in each of the training, validation and test segments. The dataset qualifies only if at least $S_{\text{min}} = 6$ satellites are retained spanning at least $R_{\text{min}} = 3$ distinct orbital regimes, regimes being derived from the ingested elements as altitude $\times$ inclination cells, not from acquisition intent.

Eleven histories were acquired; nine were retained (Table I) spanning five regimes with all six bands supported. Two newly launched Starlink objects were excluded because their ≈ 4.5 -day histories cannot fill a chronological split at any band—an insufficient-coverage exclusion decided by the qualification rule, not by learning performance.

B. Canonical Orbital History

Each object’s history is distributed in two encodings of the same solutions. The JSON element history is the

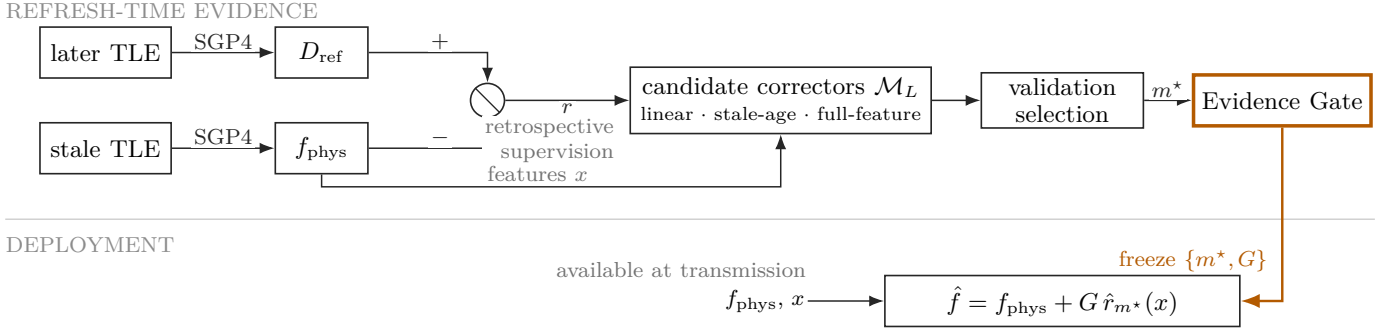


Fig. 1. Evidence-gated residual correction. Refresh time: a later element and the stale element in use are each propagated by SGP4, and the subtraction junction forms the retrospective residual target of Eq. (3); because the later element does not exist at the original transmission, r supervises only correctors applied to subsequent transmissions. With the feature vector x it drives the candidate set $\mathcal{M}_L$, from which validation selects m^* ; the Evidence Gate then compares that branch’s validation error with the physics baseline’s, $\text{MAE}_{m^*}(V) < \gamma \text{MAE}_{\text{phys}}(V)$, and freezes $\{m^*, G\}$. Deployment: the frozen state is applied without refitting or re-gating. The split, screening rule and diagnostic reference predictors are in Sec. III and Algorithm 1.

single canonical scientific input; the archival TLE file is kept for provenance and never ingested as an additional observation set, since doing so duplicates each solution and collapses the apparent update cadence toward zero. Epochs are normalised deterministically to a fixed quantum and grouped by catalogue number and normalised epoch.

Such a group is not automatically a duplicate: SpaceTrack republishes an epoch when the solution is re-fitted, and 884 groups here hold two or more physically distinct solutions. Equivalent duplicate collapse covers only rows agreeing on mean elements, drag term and ephemeris type and differing in publication bookkeeping; the rest need same-epoch revision resolution, which selects one solution by publication metadata, never by averaging.

The canonical rule is the earliest published revision—an availability argument, not a performance one. In 237 of the 884 groups the last revision was created only after the next element set already existed, so choosing it would let a terminal correct with a solution that did not yet exist. Canonicalisation leaves 46 859 elements from 52 538 raw rows for the largest object and restores plausible update cadences (Table I). The retained identifier, creation date and competing-revision count are recorded for every resolved epoch.

Canonicalisation is not load-bearing: resolving the same groups to the latest revision leaves every headline count unchanged. Deleting all contested epochs—an ablation, not a rule, since it discards real observations—opens one configuration, BLACK KITE-2 at 48 h, whose validation gain rises to 6.1% while its held-out result is -4.3% on 37 pairs, with Holm-adjusted $p = 0.52$. That is the validation/test mismatch this paper is about, not deployable learnability; it is the hollow marker in Fig. 2.

C. Pair Construction and Staleness Bands

For each reference (newer) element set we select the older set whose epoch gap is closest to the staleness target within an admissible band. The six targets and their bands

[lo, hi] in hours are 8 : [4, 14], 24 : [16, 36], 48 : [36, 60], 72 : [60, 84], 96 : [84, 120] and 168 : [144, 192]. Consecutive element sets are not required: this reproduces the case in which a terminal holds a TLE roughly τ hours old. The pair is propagated open-loop and compared to the reference at $K = 24$ UTC samples spanning one orbital period (5700 s), at an 868 MHz carrier for a representative ground station (24°N, 121°E). A pair is screened out if any of its samples exceeds $\max |r| > \tau_s$, with $\tau_s = 1500$ Hz as the pre-specified value. Screening is applied before the chronological split, so it changes the population that is split rather than the split boundaries; Sec. IV-C varies τ_s explicitly. Splits are 60/20/20 by reference epoch, using two global boundaries shared by every band of an object.

D. Residual Candidate Models

The feature vector x_t has ten entries: TLE age and epoch gap, the stale-TLE Doppler, sin and cos of the orbital phase, elevation and slant range, and the stale mean motion, drag term and eccentricity. The learned candidate set $\mathcal{M}_L$ holds three fitted correctors; two further predictors are diagnostic references and never enter selection:

$$\mathcal{M}_L : \hat{r}(x_t) = a t_{\text{age}} + b, \quad \hat{r}(x_t) = w^\top z(t_{\text{age}}, t_{\text{gap}}) + w_0, \quad (5)$$

$$\text{ref.: } \hat{r} = 0 \text{ (SGP4)}, \quad \hat{r} = c, \quad c \in \{\text{mean}(r_T), \text{median}(r_T)\}$$

with $z(\cdot)$ standardisation fitted on training and the ridge penalty chosen from $\{10^{-2}, \dots, 10^3\}$ on validation. Selection runs over $\mathcal{M}_L$ only, $m^* = \arg \min_{m \in \mathcal{M}_L} \text{MAE}_m(V)$: the zero predictor is the physics baseline the gate compares against and the constant-bias predictors are diagnostic, so neither competes for m^* nor can open the gate.

$\mathcal{M}_L$ is deliberately low-capacity: the question is whether low-dimensional, chronologically stable residual structure exists at all, a precondition for high-capacity learning to be worth its deployment and audit burden. Its low

capacity limits, rather than eliminates, validation-window overfitting.

E. Statistical and Deployment Evaluation

The independent experimental unit is the accepted TLE pair, never the individual propagation sample: the $K = 24$ samples within a pair are one propagation over one orbit and are strongly correlated. Every metric is computed per pair then averaged, and every statistic is paired at pair level. We report the pair-level MAE, the learned win fraction, an exact two-sided sign test, a bootstrap interval resampling pair identifiers, a block bootstrap resampling whole reference-epoch days to respect dependence between adjacent pairs, and Holm-adjusted p -values across the family of tests.

These statistics are diagnostics and never modify the deployment decision: G depends only on the margin test of Eq. (4), so a result can be strongly significant and still refused, and a marginal one is never promoted for a small p -value. Keeping the two separate is the point of Eq. (1).

Before any result is emitted the pipeline asserts mechanically that train, validation and test pair identifiers are disjoint, that validation strictly precedes test, that one accepted pair contributes exactly one observation, and that G is reproducible from validation quantities alone.

Endpoint mapping. To convert a residual into terminal cost we use control proxies: a guard proxy $g = 2p_{99}(|e|)$ on the error e of whichever branch the gate selects, an outage proxy $\rho = \Pr[|e| > F_{\text{tol}}]$ with a representative sub-kHz tolerance $F_{\text{tol}} = 500$ Hz—a modelling choice, not an LR-FHSS conformance requirement—and, treating timing and frequency misses as independent, a joint success proxy and energy per burst

$$S = (1 - P_t)(1 - P_f), \quad E_{\text{succ}} = \frac{I_{\text{rx}}V(g_t + t_{\text{rx}}) + P_{\text{tx}}t_{\text{tx}}}{S}, \quad (6)$$

with $P_f = \text{erfc}(F_{\text{tol}}/(\sigma_{\text{res}}\sqrt{2}))$, σ_{res} the held-out residual standard deviation, $g_t = g_0 + 3\sigma_t$ the timing guard and σ_t mapped from TLE age. Parameters follow the analysis scripts ($g_0 = 30$ ms, $t_{\text{rx}} = 50$ ms at 12 mA/3.3 V, 14 dBm for $t_{\text{tx}} = 200$ ms). These are control-layer proxies for margin sizing, not packet-error, bit-error or link-budget measurements.

IV. Results

A. Multi-Satellite Deployment Result

Figure 2 reports the 54 target-specific configurations. No learned correction earns deployment at the pre-specified operating point: the gate is closed in every object and band, and the limited BLACK KITE cross-direction checks are closed in all 12. We do not claim a complete cross-satellite transfer matrix.

The held-out consequences are heterogeneous even though the decision is not. SENTINEL-6B degrades by 8–31% at every band and ONEWEB-0015 by up to 122%, while ISS moves by less than 0.4% anywhere and both

TABLE I

The nine retained objects. Elements are canonical solutions after equivalent-duplicate collapse and same-epoch revision resolution; all six staleness bands are supported for every object. Eleven histories were acquired and two were excluded for insufficient chronological coverage. All values are model-derived from element sets, not measured.

Object	Regime (alt. / incl.)	Alt. [km]	Incl. [°]	Elem.	Gap [h]
ISS (ZARYA)	very low / mid-incl.	388	51.6	46 859	4.52
FLOCK 4H 1	low / SSO	507	97.4	614	7.89
FLOCK 4H 2	low / SSO	507	97.4	635	7.89
BLACK KITE-1	low / SSO	507	97.4	446	6.33
BLACK KITE-2	low / SSO	590	97.8	252	8.04
IRIDIUM 177	mid / near-polar	629	86.2	2 623	9.73
IRIDIUM 181	high / near-polar	743	86.6	2 678	9.98
ONEWEB-0015	upper / near-polar	1114	87.7	1 695	8.00
SENTINEL-6B	upper / mid-incl.	1336	66.0	727	5.62

IRIDIUM objects improve slightly at several bands. The refusal is uniform not because harm is, but because no branch clears the validation margin.

The screened fraction grows with staleness, from under 0.3% at 8 h to 28.5% for ISS and 16.9% for ONEWEB-0015 at 168 h, as longer propagation exposes manoeuvre and bad orbit-determination epochs. Effect size and resolution are also misaligned: ISS contributes 14 000–19 000 held-out pairs per band, so its 0.1% degradations are overwhelmingly significant, while SENTINEL-6B reaches 31% on 131 pairs. The gate, a margin rather than a p -value, is unaffected.

B. Detectable but Sub-Margin Residual Structure

A refusal is only informative if the learner can find structure where it exists. It can. For IRIDIUM 181 at 8 h the validation-selected corrector reduces held-out pair-level MAE from 0.16740 to 0.16414 Hz, a 1.94% improvement winning on 61.8% of 633 pairs (Fig. 3(a)). It survives Holm correction and a day-blocked bootstrap, has a bootstrap 95% interval of $[-4.1, -2.4]$ mHz excluding zero, and—unlike most signals here—keeps its sign at every screen (Fig. 3(b)). Comparable sub-margin gains appear for IRIDIUM 177 at 96 and 168 h, and the stale-age ridge is selected in every such cell, consistent with the correction tracking element staleness rather than pass geometry.

The gate nonetheless stays closed: the validation gain is 1.37% against the 5% that $\gamma = 0.95$ requires—a deployment margin working as intended, not a missed detection. Reporting only the relative gain, as a prediction-accuracy study would, would have made this cell look deployable.

C. Screening Robustness

Because the residual screen removes outlier pairs, it could in principle manufacture either conclusion. We re-executed the pipeline independently at none, 150, 500, 1500 and 3000 Hz, refitting and re-gating at each: 270 configurations.

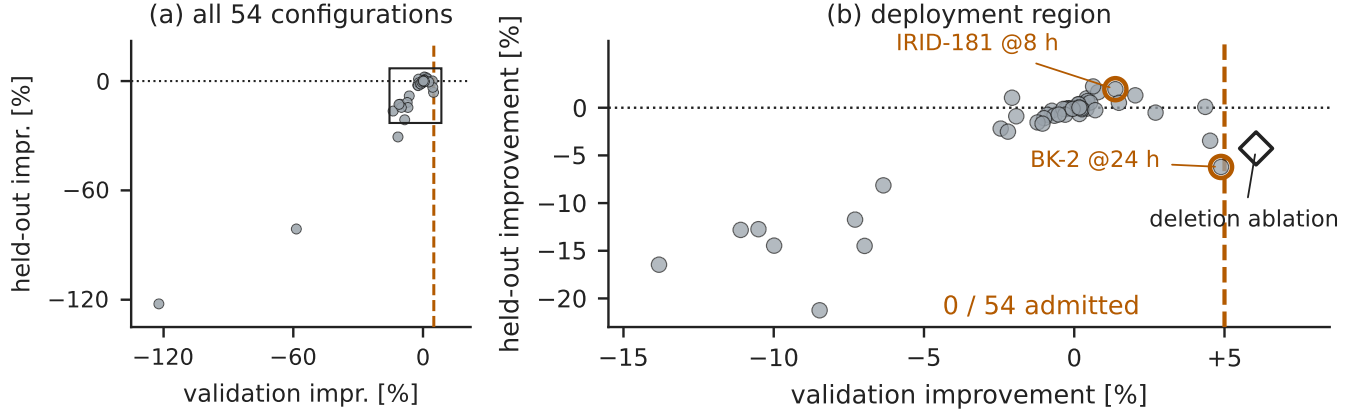


Fig. 2. What validation told the gate, and what happened next, for all 54 target-specific configurations at the pre-specified operating point ($\gamma = 0.95$, 1500 Hz screen). Positive means the learned branch is better on both axes. (a) All configurations at true coordinates; outcomes are strongly heterogeneous, ONEWEB-0015 at 24 h reaching -122% on both axes, and the rectangle marks the region enlarged in (b). (b) The gate reads only the horizontal axis and admits a branch beyond the dashed $+5\%$ margin, which 0 of 54 reach; the nearest miss is BLACK KITE-2 at 24 h ($+4.9\%$ validation, -6.2% held-out). The hollow diamond is the non-canonical conflict-deletion ablation of Sec. III-B, the only configuration that crosses the margin under any canonicalisation setting tested, and harmful on held-out data when it does. Cross-direction continuity opens 0 of 12; residuals are model-derived inter-TLE differences, not measured Doppler.

Exactly one of the 270 opens: IRIDIUM 177 at 168 h under the 150 Hz screen, which discards 40.3% of candidate pairs. Adjacent thresholds contradict it, its Holm-adjusted p is 1.00, and it is not the pre-specified operating point (Fig. 3(b)).

Comparing no screening against the pre-specified rule across all 54 cells, screening improves the learned branch’s relative performance in 36, worsens it in 4 and leaves 14 unchanged—all cells where it removes no pairs—with a median shift of 1.26 percentage points in its favour. The aggregate direction is inconsistent with the 1500 Hz screen manufacturing the negative result; training-composition and evaluation-membership effects are not causally separated.

D. Deployment Margin versus Endpoint Value

No cell in Fig. 2 produces a resolvable endpoint change, because the decisive comparison is relative against absolute scale: the IRIDIUM 181 improvement is 1.94% relative but 3.26 mHz absolute, against $F_{\text{tol}} = 500$ Hz. With $\sigma_{\text{res}} = 0.545$ Hz the tolerance lies 917 residual standard deviations away, so the frequency-miss term of Eq. (6) is $\log_{10} P_f < -1.8 \times 10^5$ for both branches. The guard proxy moves from 4.470 to 4.439 Hz, 2.3×10^{-7} of the hopping bandwidth; the outage proxy is zero for both; and S and E_{succ} are indistinguishable at double precision, both 15.94 mJ per burst (Fig. 3(c)). A 30.7% degradation for SENTINEL-6B is likewise endpoint-invisible, which is why relative MAE alone is a poor indicator of endpoint relevance.

Margin sensitivity. Recomputing the gate at $\gamma \in \{1.00, 0.99, 0.98, 0.975, 0.95, 0.90\}$ gives 24, 7, 5, 4, 0, 0 openings, of which 16, 4, 2, 1, $-$, $-$ improve on held-out data. Tightening the margin does not monotonically

improve held-out selection precision, which falls from 66.7% to 25.0% before openings vanish; what falls monotonically is the number of harmful deployments, 8 to 0.

V. Discussion, Limitations and Conclusion

The substantive finding is the separation in Eq. (1): residual structure is detectable—the IRIDIUM 181 result survives multiplicity, temporal dependence and screening variation—yet it is sub-margin and leaves the endpoint budget unchanged. A policy deploying on detectability alone would have accepted a branch with no resolvable benefit, its deployment burden and its exposure to distribution shift.

Design implications. Three consequences follow. First, statistical significance is not the deployment criterion: the IRIDIUM 181 effect is significant under every test we applied and is still refused, because deployment turns on the margin a branch clears before commitment, not the p -value it attains afterwards. Second, the durable signal is selected by the stale-age ridge wherever it appears, pointing to a low-dimensional ageing bias rather than unexploited pass geometry; but its absolute size—3.26 mHz against a 0.545 Hz residual spread—does not justify higher-capacity models. Third, the deletion ablation exposes the gate’s limit: contested-epoch removal lifts one BLACK KITE-2 cell across γ on validation, yet it loses 4.3% held out. The margin is an evidential filter, not a guarantee of held-out benefit.

Limitations. All results are software-only and model-derived: every reference is a later element solution propagated by SGP4, not measured Doppler, and no live-satellite contact, LR-FHSS decoding, error-rate or gateway-acknowledgement result is claimed. The endpoint

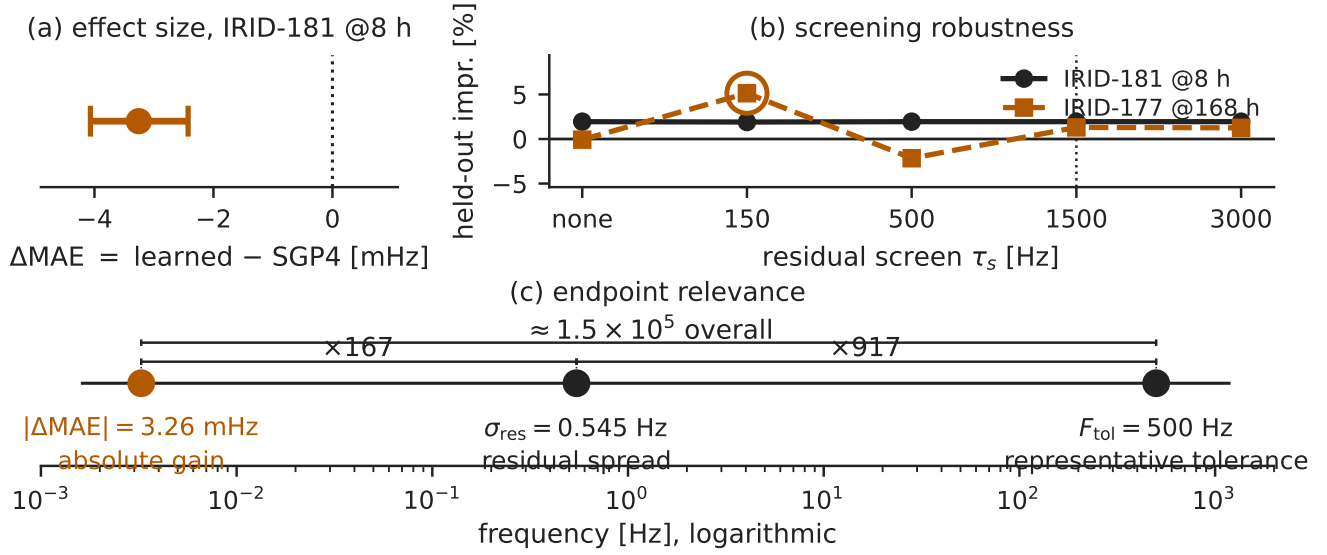


Fig. 3. Three criteria on the same effect. (a) Pair-level effect size for IRIDIUM 181 at 8 h: point estimate and 95% pair-bootstrap interval over $n = 633$ held-out pairs, with the learned branch better on 61.8% of them; the effect survives Holm correction and a day-blocked bootstrap. (b) That sign holds at every screening threshold, whereas the single IRIDIUM 177 opening at 150 Hz (hollow marker; the only opening among 270 configurations, Holm-adjusted $p = 1.00$) is contradicted by its neighbours; the dotted line is the pre-specified screen. (c) The same gain against the endpoint scale. The three quantities are not interchangeable: they are the absolute size of the correction, the spread of the residual it acts on, and a representative control tolerance. All endpoint quantities are software control proxies.

budget is a control proxy depending on the representative $F_{\text{tol}} = 500 \text{ Hz}$, though the relative statement survives plausible re-choices. Nine objects do not represent all LEO spacecraft, the cross-direction checks are a continuity test rather than a transfer matrix, and screening effects on training and evaluation membership were not decomposed. One feature, the stale-to-reference epoch gap, is fixed by the reference element's publication time and is therefore not computable at the transmission epoch; two of the three candidates consume it, so the selections reported here characterise attainable residual structure rather than a branch a terminal could evaluate unaided. Validation is reused for ridge-penalty tuning, candidate selection and the gate test, so the margin is selection-conditioned and that reuse favours the learned branch; the held-out segment is touched by none of the three. Packet-level and then authorized over-the-air measurement remain future work.

Reproducibility. Qualification, canonicalisation, the pair-level protocol, all 54 primary and 270 screening configurations, the γ frontier and the endpoint proxies are produced by version-controlled scripts from committed artifacts, with raw archives held outside the repository under a per-object SHA256 manifest. Figures are rendered from those frozen artifacts.

Conclusion. A validation-gated endpoint policy retains the SGP4 baseline everywhere at the pre-specified operating point, and the most robust detectable residual correction is worth 3.3 mHz on a budget that cannot resolve it. The contribution is not that learning fails, but that three questions usually collapsed into one—is the

gain real, is it large enough, is it worth anything at the endpoint—have different answers on the same data, and that a gate can ask the second before paying for the third.

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